

IBM Resiliency Summit

August 11-13, 2026



Resiliency Testing

Testing for Real-World Failures: Validate your resilience strategy through rigorous testing practices that simulate real-world disruptions

Day 2 - 4:05 PM – 4:50 PM

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- 03 AMEX Testimonial
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- 05 The Cost of Not Testing
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- 06 Shift Left on IBM Z
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- 07 Architectural Guiding Principles
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- 08 TTEE Framework
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- 10 IV&V — Independent Verification & Validation
Correctness · Quality · Compliance

Bottom Line Up Front

The Compelling Need for Testing on IBM Z

- **Risk creates the need.**
 - IBM Z runs the world's most critical workloads — 68% of global transaction value, 92 of the world's top 100 banks, 23 of the 25 largest airlines.
 - A single hour of mainframe downtime can cost enterprises over \$600,000 to \$1,200,000 in lost revenue and damages —for 91% of large enterprises.
- **The exposure grows when weaknesses are discovered late:** Defects cost 15× more to fix in system test than in development, and up to 100× more if they reach production.
- Yet most Z organizations still rely heavily on manual, ad-hoc testing — creating compounding risk.
- Testing is not a cost center — it is a resiliency investment that directly protects revenue, regulatory standing, and brand trust.
- The organizations that test systematically, continuously, and independently are the ones that recover fastest and fail least.

Testing on IBM Z is not optional — it is a strategic imperative.

IBM High Availability Center of Competency (HACoC)



Our mission is to enable clients to achieve their resiliency goals and maximize the value they receive from their IT infrastructure.

Our unique value to clients

- Focus on how IT infrastructure impacts application service availability to the business
- Review how the technology is implemented and how applications use the infrastructure to deliver the required service
- Extensive knowledge of industry trends, availability challenges and solutions
- Proven track record of success with hundreds of clients across every industry
- Deep expertise in Availability and Service Management – participation in IT Service and Operational Management communities
- Regulatory mandates such as DORA and NIS2 as a catalyst for enhanced business outcomes through increased operational resiliency
- Engagement teams that tap IBM's deep bench of IT resiliency expertise, in both technology and service management disciplines

Proven, flexible methodology

Workshops can be conducted in-person, hybrid and remotely.



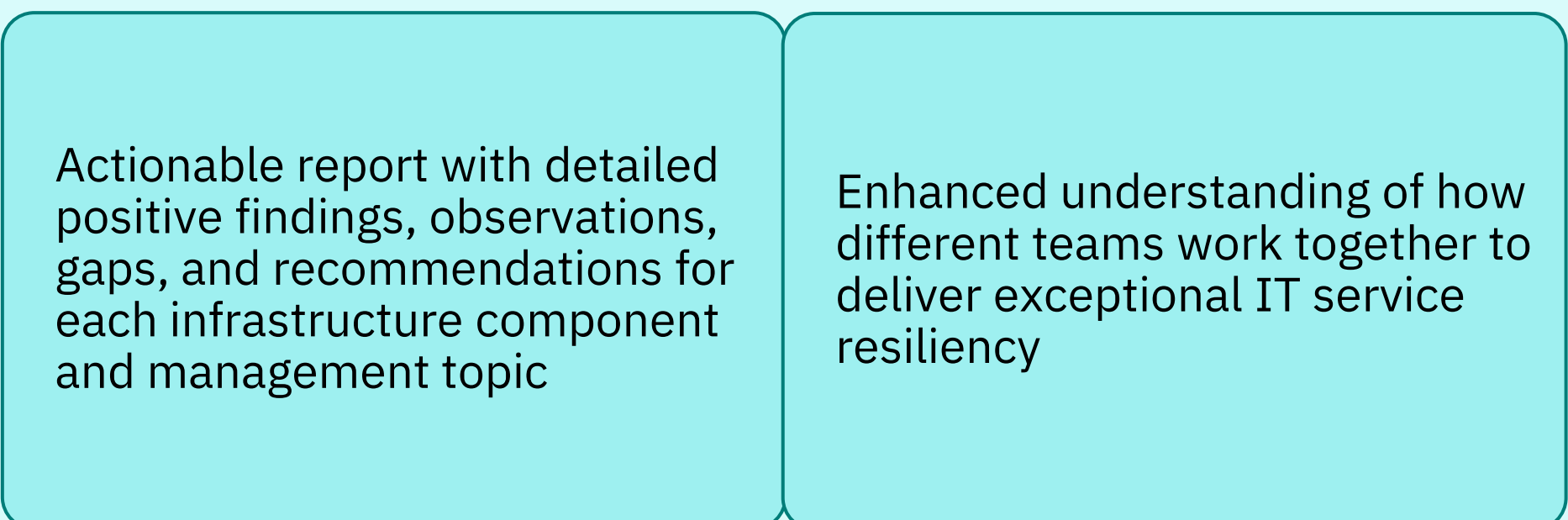
Extensive Capabilities

Tailored engagement for each client's specific IT resiliency objectives.

<i>Availability workshop</i>	<i>Post-incident review</i>	<i>Expert support</i>
In-depth technology and service management review of key IT infrastructure	Analyze recent outage to identify root and all contributing causes	Support client-led meetings Support other IBM teams Share HA knowledge

What we deliver

Compelling mix of partnership, collaboration, and expertise.



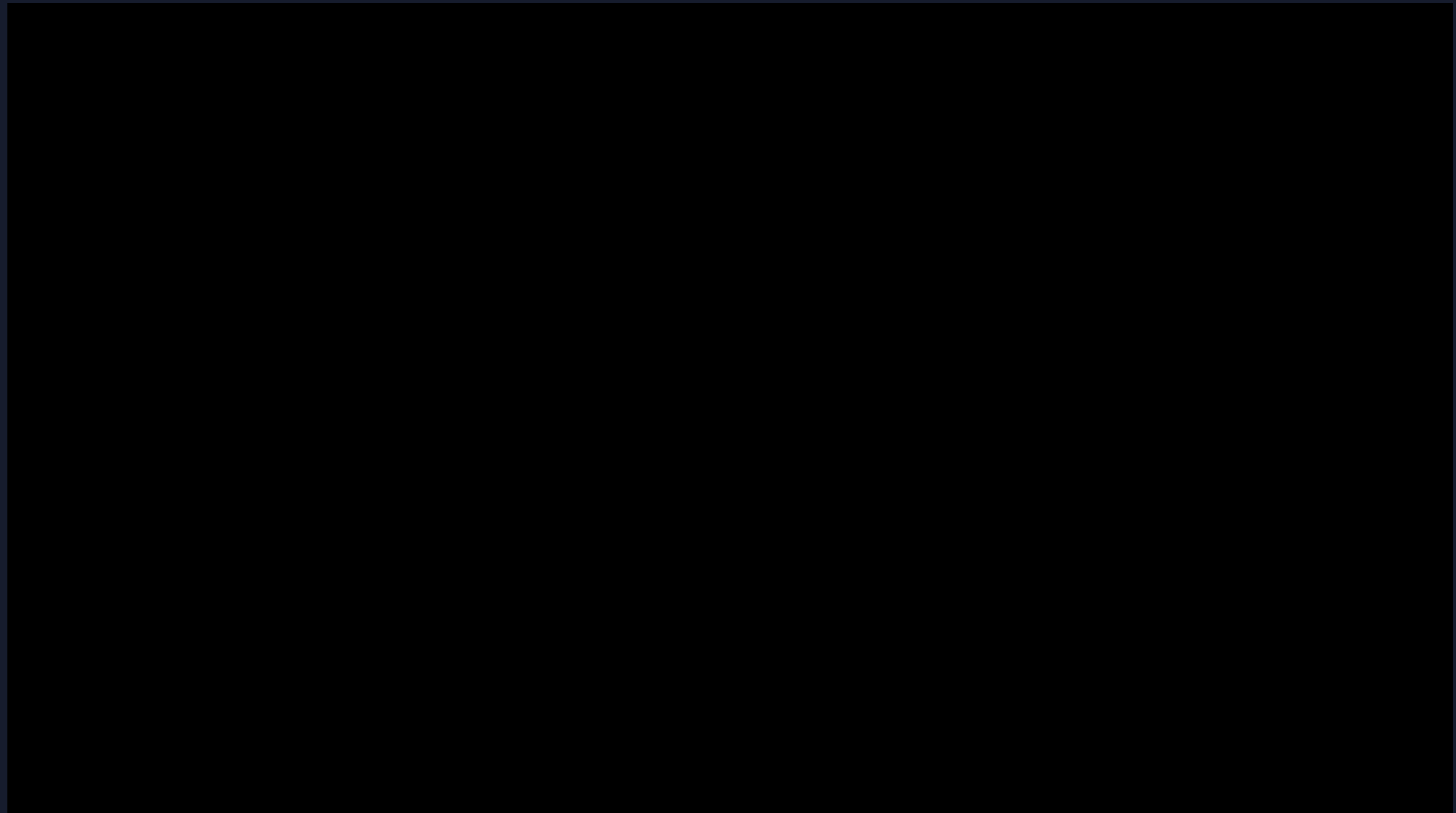
AMEX Testimonial

Real-World Z Testing in Action

Brian Callaghan

VP Global Infrastructure Services / Mainframe, American Express

American Express operates one of the world's most demanding IBM Z environments — processing billions of transactions annually. Their testing program is a model for the industry.



The Big Three of Testing from a Z Perspective

1. Infrastructure Testing

- ❑ Validates hardware, firmware, and system software integrity
- ❑ Sysplex coupling, I/O paths, network connectivity
- ❑ Planned and unplanned failover validation
- ❑ Performance and capacity baseline verification
- ❑ Confirms Reliability and Redundancy tiers of the resiliency model

2. Service Operations Readiness Testing (SORT)

- ❑ Validates operational readiness before go live
- ❑ Transaction test scripts executed against service levels
- ❑ Monitors response time, throughput, and error rates
- ❑ Five-phase cycle: Plan → Script → Execute → Analyze → Tune
- ❑ Integrated into Enterprise Engineering Test Management (ETM)

3. Validation of Safeguarded Copies

- ❑ Confirms that cyber-resilient backup copies are recoverable
- ❑ Tests Safeguarded Copy / FlashCopy / GM replica integrity
- ❑ Validates RTO and RPO against defined service levels
- ❑ Detects silent corruption before a crisis demands recovery
- ❑ Critical for DORA, SOX, and regulatory compliance

The Cost of Not Testing

Manual Testing ROI & Hidden Risk Exposure

Example ROI calculation...

- Developer cost baseline: at \$50/hr with 30% of time on manual testing, each developer costs \$31,200/yr in test labor alone.
- For an enterprise with 500 COBOL developers, total annual manual testing cost = \$15.6M.
- IBM Test Accelerator for Z (TAZ) delivers 20% Year-1 savings = \$3.12M annual reduction.

Hidden costs multiply the exposure:

- Repetitive, inefficient, and non-reusable manual test cycles waste developer time that could drive innovation.
- Undetected defects reaching production cost 100× more to fix than defects caught in development.
- Operational incidents from untested code average >\$300K per hour of downtime.
- Mainframe CPU cycles consumed by manual test runs are not available for revenue-generating workloads.
- Regulatory penalties for inadequate testing controls (DORA, SOX, PCI-DSS) can dwarf tool investment costs.

The question is not whether you can afford testing, it's whether you can afford failure.

IBM Test Accelerator for Z (TAZ)

Core Capabilities

- ❑ ZUnit: COBOL/PL/I unit testing with record & replay — no middleware required
- ❑ ZVTP (Z Virtual Test Platform): Integration test via batch replay of CICS/IMS transactions
- ❑ AI-assisted test creation: Natural language and UI-driven functional test case generation
- ❑ Virtual z/OS sandboxes: Spin up isolated z/OS environments on Linux on Z or Intel x86
- ❑ Code coverage acceleration: Identify untested paths; generate test data for conditional branches
- ❑ IDE integration: Eclipse and VS Code; Git-native test case storage
- ❑ Pipeline integration: Jenkins, GitLab, DBB — tests run on every commit

Competitive Differentiators

- ❑ Only vendor with virtual z/OS sandbox environments
- ❑ AI-powered test creation — NL and UI driven
- ❑ Multiple test types in a single license: unit + functional + integration
- ❑ Broadest subsystem support: CICS, IMS, Db2, batch, APIs
- ❑ Tightest IDE integration across Eclipse and VS Code
- ❑ Part of the broader IBM ELM/ETM ecosystem for end-to-end traceability

Shift Left: Continuous Testing on IBM Z

Testing Early, Often, and Automatically

- Unit tests are fast, isolated, and cheap; they should form the base of automation.
- Unit Testing (ZUnit): Developer-written tests for individual COBOL/PL/I programs — catch defects before code is committed.
- Integration Testing (ZVTP): Tests transaction flows and program-to-program interfaces in batch — no live CICS environment needed.
- Functional / System Testing: End-to-end validation of application behavior across subsystems (Db2, CICS, IMS).
- CI/CD Pipeline Integration: Jenkins, GitLab, DBB orchestrate automated build → test → deploy for every code change.
- IBM Test Accelerator for Z (TAZ): AI-assisted test creation, virtual z/OS sandboxes, code coverage acceleration.
- Key metric: Organizations with mature Shift Left practices detect 5× more defects before integration test, reducing overall cycle time by 30–50%.

Shift Left = test earlier, fix cheaper, release faster.

Architectural Guiding Principles

Non-Production Z Environment Design

- Non-production environments must mirror production architecture — including Sysplex coupling, storage configuration, and network topology.
- Dedicated non-prod capacity prevents test workloads from competing with production — protecting SLAs and test result fidelity.
- Environment parity: Test what you run in production. Configuration drift between non-prod and prod is a leading cause of missed defects.
- Security parity: Apply the same security controls (RACF, encryption, audit) in test environments — safeguarded copy validation requires it.
- Data management: Use masked, production-like datasets (IBM Optim). Never use live production data in non-prod environments.
- Version control: Store test cases, test data configurations, and environment definitions in Git alongside application code.
- Capacity planning: Ensure non-prod can absorb SORT workloads, safeguarded copy validation, and CI/CD pipeline test runs concurrently.
- Recovery testing environments need isolation — a botched DR test must not contaminate production or other test streams.

Non-Production Z: More Than Just Testing

TTEE — Test · Train · Exercise · Experiment

TESTING

- Demonstrates correct operation of equipment, procedures & systems
- Alert & notification systems
- Backup infrastructure (power, storage, network)
- Recovery & failover procedures
- Accessibility and interoperability

TRAINING

- Familiarizes staff with roles during disruption
- Continuity plan activation vs. normal operations
- Backup communications and IT systems
- Orders of succession & delegations of authority
- ETM onboarding and test process adoption

EXERCISING

- Validates plans in low-risk simulated scenarios
- Enterprise-wide backup capabilities
- Recovery from continuity plan activation
- Virtual z/OS sandboxes — fail safely, learn fast
- Transition back to normal operations

EXPERIMENTATION

- Safe exploration of new configurations and tools
- Chaos engineering on Z — inject failures, observe behavior
- AI-driven anomaly detection validation (TDz)
- New workload onboarding before production promotion
- Continuous improvement planning

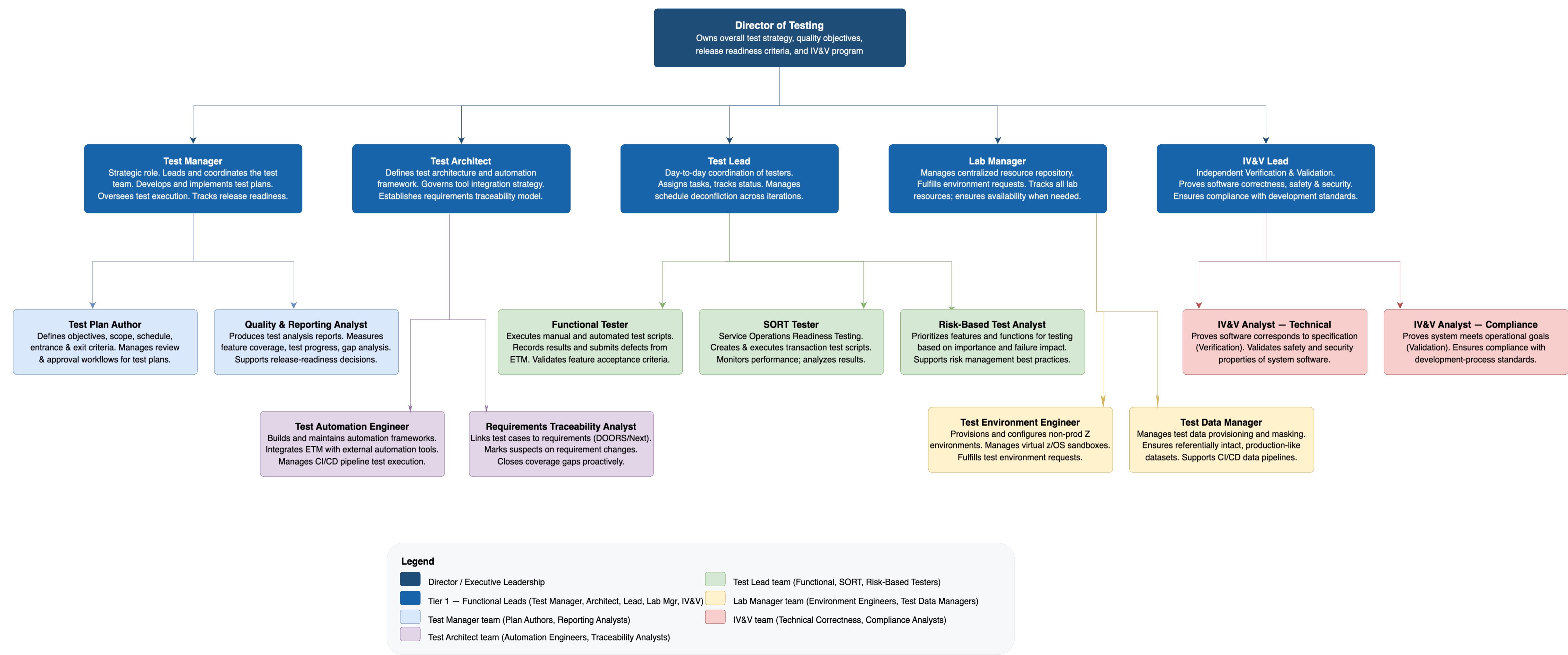
A Testing Discipline is Necessary

Organizational Structure for Z Testing

- A formal testing organization is not overhead — it is infrastructure for resiliency. Without it, testing is ad hoc, inconsistent, and unreliable.
- Recommended roles (from IBM Engineering Test Management):
 - Test Manager: Strategic leader. Develops test plans, oversees execution, owns release-readiness decisions.
 - Test Architect: Defines automation frameworks, tool integration strategy, and requirements traceability model.
 - Test Lead: Day-to-day coordination, iteration scheduling, deconfliction across teams and environments.
 - Functional Tester / SORT Tester: Executes scripts, records results, submits defects through ETM.
 - Lab Manager: Manages non-prod Z environments, fulfills test environment requests, tracks all lab resources.
 - IV&V Lead: Independent Verification & Validation — separate from development, validates correctness and compliance.
- Outside the test org: Requirements owners, developers, business stakeholders.

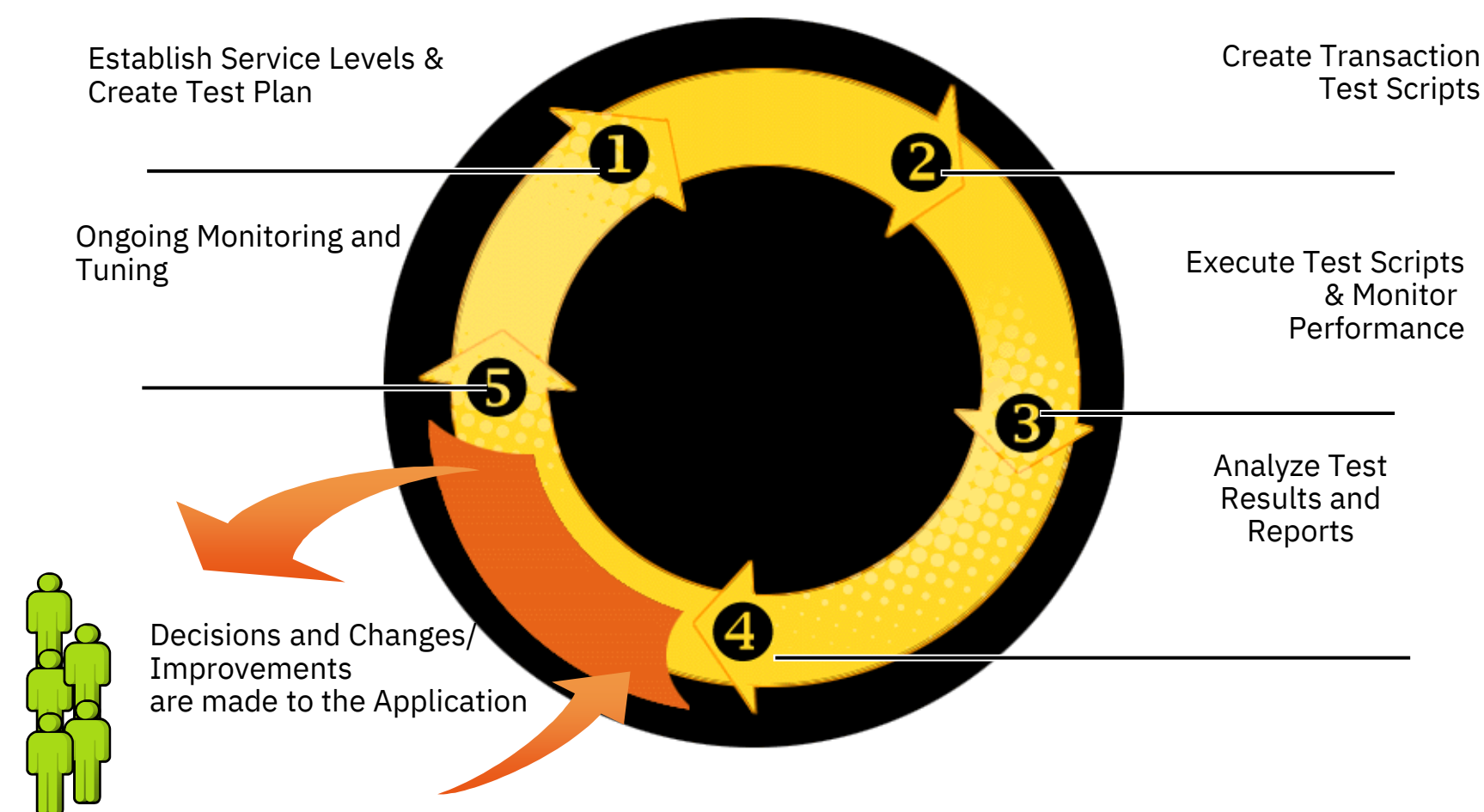
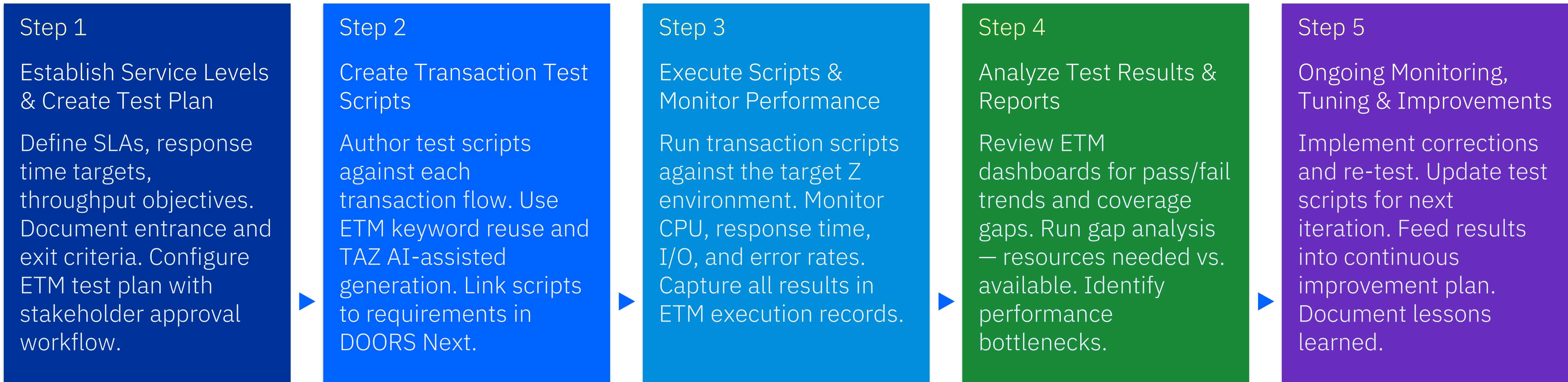
Testing Organization

Organizational Structure for Enterprise Testing



Service Operations Readiness Testing (SORT)

Five-Phase Methodology within IBM Engineering Test Management



Administration & Scheduling Deconfliction

Managing Competing Demands on Non-Production Z

- Non-production Z environments are shared, constrained resources — contention is the #1 cause of test delays.
- ETM lab management: centralized resource repository tracks all lab assets and fulfills environment requests systematically.
- Scheduling governance: Test Lead owns the master schedule; all SORT, CI/CD, DR validation, and safeguarded copy tests are coordinated.
- Environment reservation: testers request, lab manager approves and reserves machines — preventing conflicts and idle time.
- CI/CD isolation: pipeline-triggered automated tests run in dedicated TAZ virtual sandbox environments — never on shared LPAR.
- DR test windows: scheduled independently from functional test cycles; protected from CI/CD noise.
- Change management integration: ETM is linked to EWM (Engineering Workflow Management) — test plans reflect code freeze and release milestones.
- Capacity modeling: quarterly review of non-prod utilization ensures adequate headroom for all TTEE activities.

Testing and validation to ensure you can recover from your backups

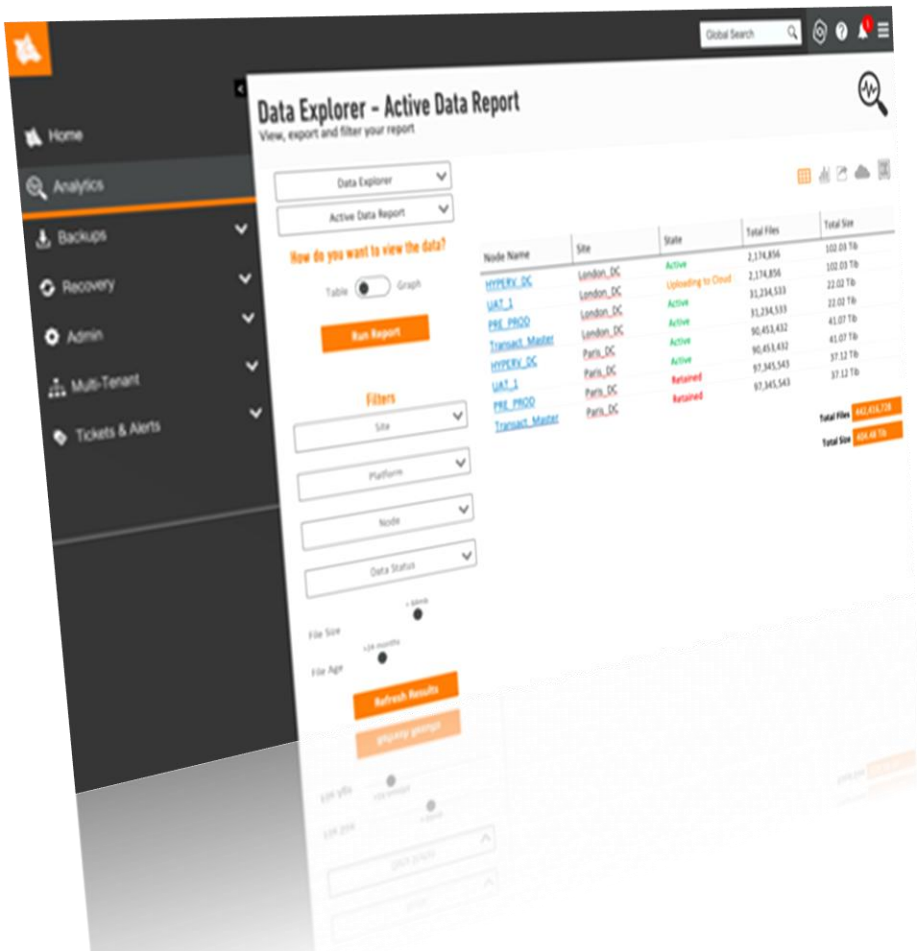
Analytics & AI



1

Analytics and AI Provide deep insight into the status of Spectrum Protect

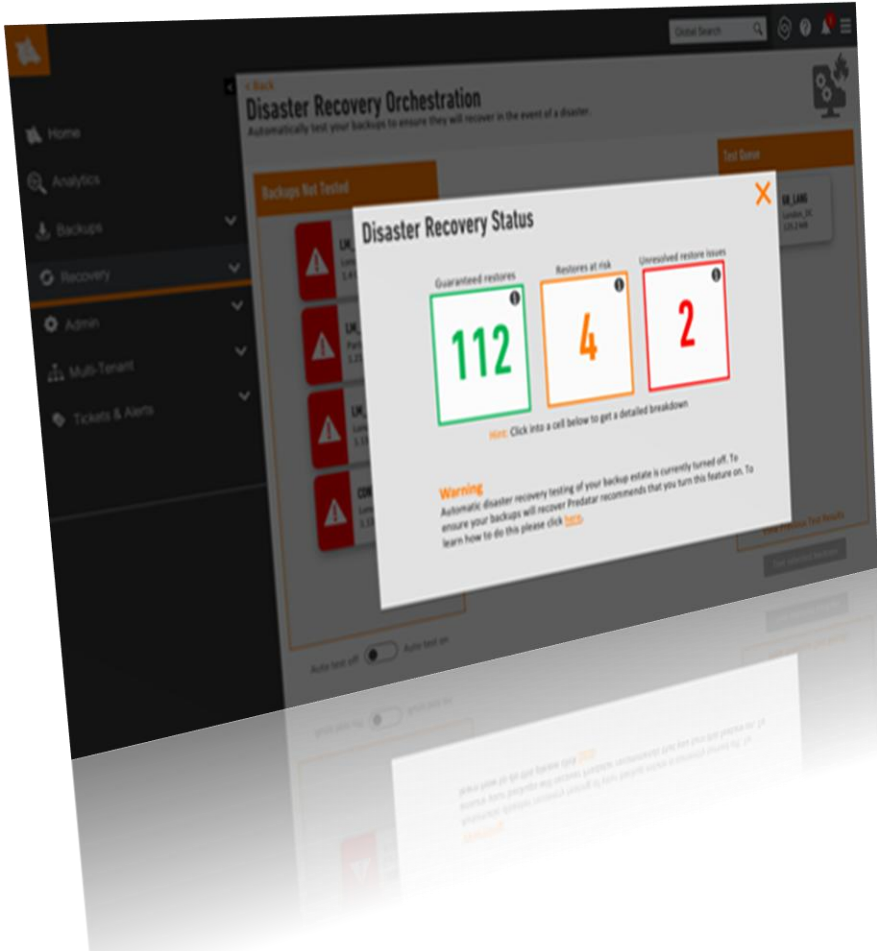
Automation



2

Automate Administration and improve operational efficiency

Recovery Orchestration and Testing



3

Orchestrate, Test and Automate Recovery

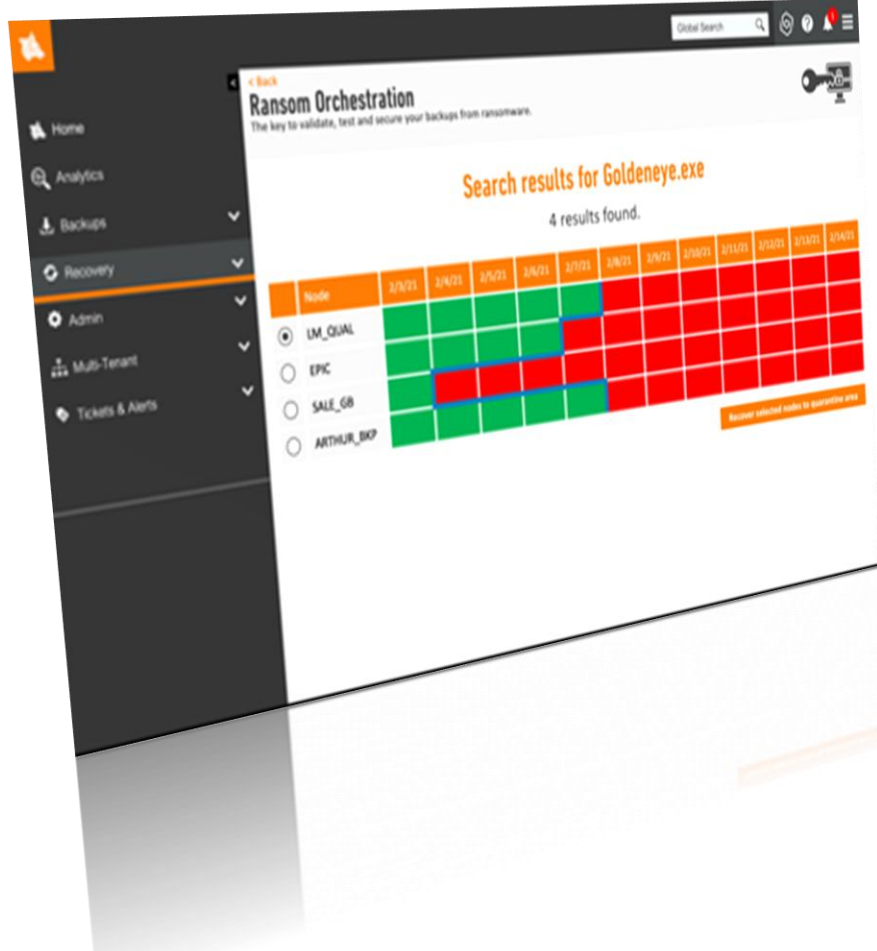
Ransomware Scanning



4

Scan for Ransomware Infections

Ransomware Recovery



5

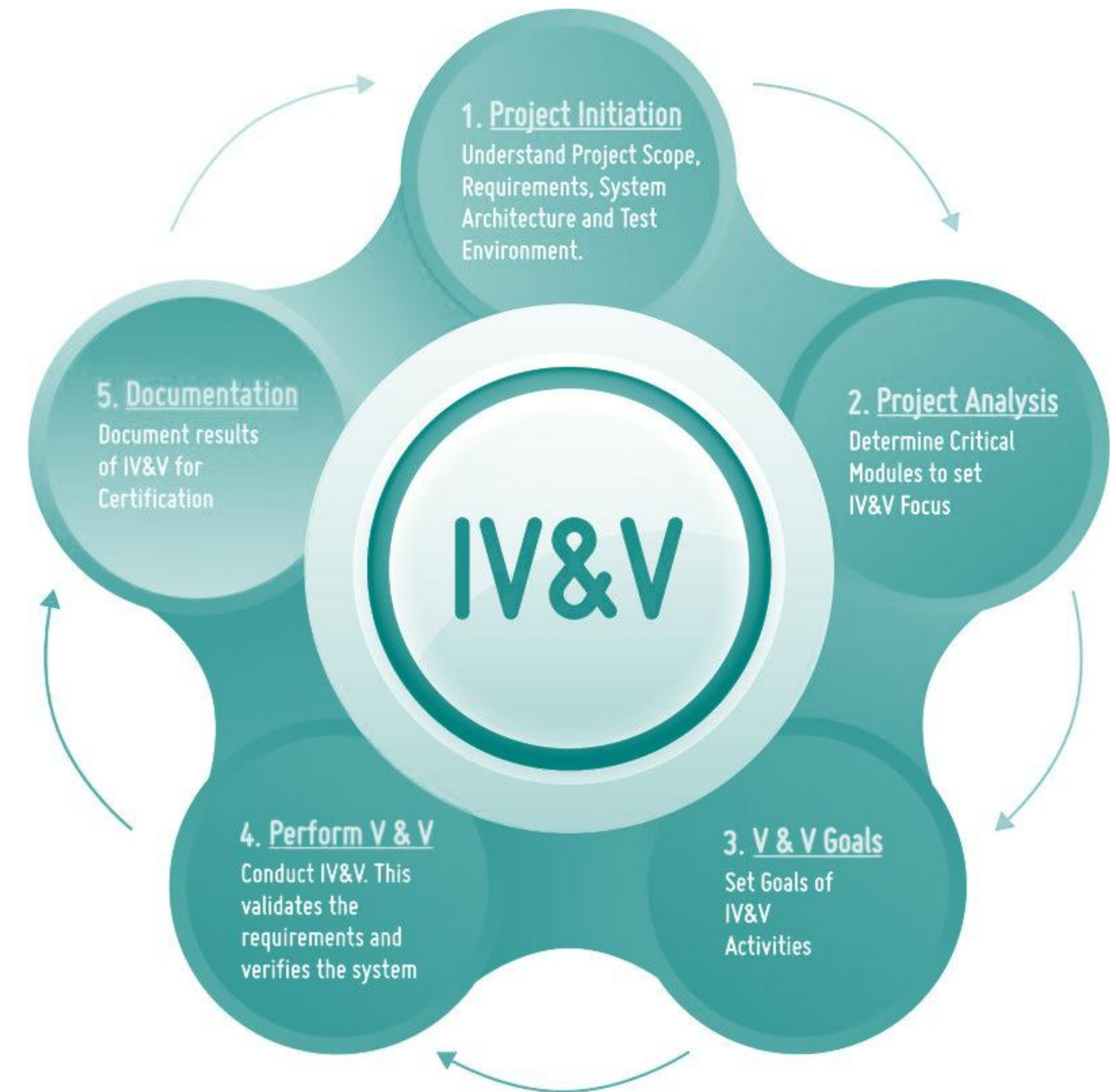
Quarantine, Clean and Recover Infected Systems

A regular cadence of validation of a clean copy is critical to your Cyber Vault success.

Independent Verification & Validation (IV&V)

Trust Through Independence

- IV&V separates the team that builds from the team that validates — eliminating confirmation bias and blind spots.
- Three primary IV&V objectives (IEEE/ISO standard):
 - Technical correctness: Demonstrates safety, security, and functional correctness of the system/software.
 - Overall quality assessment: Evaluates the complete set of software products against quality criteria.
 - Process compliance: Ensures development-process standards are followed — critical for SOX, HIPAA, DORA, PCI-DSS.
- IBM ETM supports IV&V with e-signature workflows, review/approval audit trails, and compliance-grade reporting.



IV&V is not a luxury — it is a regulatory and resiliency requirement.

Five Takeaways

Testing Early, Often, and Automatically

Testing is a resiliency investment.

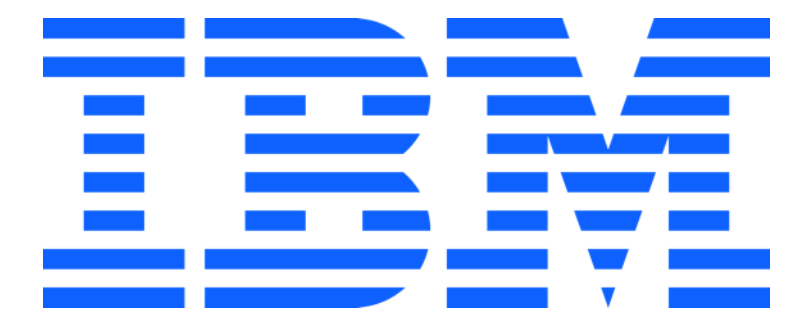
Recovery that isn't tested is theoretical.

Shift Left reduces risk and cost.

Non-production Z is a strategic asset.

Independent validation builds trust.

You don't rise to the level of your resiliency plan, you fall to the level of your testing discipline



Thank you

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